

PROJECT PROPOSAL

Route Finder & Sentiment Analysis

COURSE

Artificial Intelligence

INSTRUCTOR

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PROBLEM STATEMENT

This project implements two AI components: an **informed search algorithm** for optimal route finding between locations using real-world map data, and a **machine learning model** for sentiment analysis on social media posts. The objective is to demonstrate practical applications of A* search in navigation and classification in natural language processing, addressing real-world needs for efficient routing and understanding public opinion.

TASK 1 — INFORMED SEARCH ALGORITHM

Selected Algorithm

A* (A-Star)

A* is an informed search algorithm that finds the shortest path between nodes in a graph. It uses a heuristic function to estimate the cost from a node to the goal, ensuring optimality when the heuristic is admissible. It maintains an open set of nodes prioritised by **f-score = g-score + h-score**, expanding the node with the lowest f-score.

INITIAL STATE

Geocoded coordinates of the start location.

GOAL STATE

Geocoded coordinates of the destination location.

HEURISTIC

Haversine distance ÷ maximum speed (admissible).

TASK 2 — MACHINE LEARNING APPROACH

Task Type

Classification

Dataset Source

Reddit RSS Feed

Posts per Query

Up to 100

Algorithms/Models: TF-IDF Vectorizer for feature extraction; Logistic Regression for 3-class sentiment classification (positive / negative / neutral).

Data Preprocessing: Concatenate title + summary, remove stop words, vectorise using TF-IDF.

Training & Testing: Model trained on a small labelled dataset; applied to fetched Reddit data for live prediction.

EXPECTED OUTCOMES

Task 1	Route map with turn-by-turn directions, total distance, and estimated travel time.
Task 2	Interactive HTML table of Reddit posts with sentiment labels and confidence scores.
Metrics	Path optimality for A*; training accuracy for the ML model.

TOOLS & TECHNOLOGIES

Language	Python
Libraries	requests, folium, heapq, scikit-learn (sklearn), feedparser
APIs / Services	OSM Overpass API, Nominatim Geocoding Service
Data Structures	KD-Tree for spatial indexing

TIMELINE

TASK	EXPECTED COMPLETION
Proposal Submission	April 08, 2026
Task 1 Implementation	April 30, 2026
Task 2 Implementation	April 30, 2026
Final Report Submission	April 30, 2026

EXPECTED CHALLENGES

- API rate limits and timeouts for geocoding and OSM data fetching.
- Algorithm complexity in A* for large graphs — mitigated by KD-Tree optimisation.
- Limited training data for the ML model, potentially affecting generalisation.
- Handling one-way roads and multiple network types (drive / walk / bike).

EXPECTED RESULTS

Successful implementation of an A* route finder that computes optimal paths with interactive map visualisation and turn-by-turn directions. A sentiment analysis tool that classifies Reddit posts and presents results in a clean, interactive HTML format. Together, they demonstrate seamless integration of classical search algorithms and modern ML in practical real-world applications.

INSTRUCTOR REMARKS

Remarks:

Signature: _____